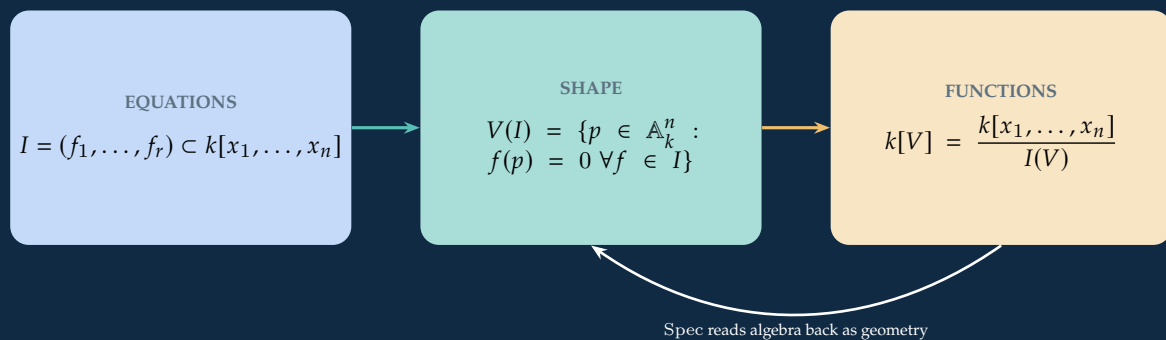


Algebraic varieties

How polynomial equations become spaces, and how geometry and algebra learn to speak to one another.



The central equivalence

A variety is a **zero set cut out by polynomials**. Its coordinate ring records the **polynomial functions that survive on that zero set**. Algebraic geometry gains its force by moving through this dictionary in both directions.

Equations become spaces

Start with affine space $\mathbb{A}_k^n = k^n$. A polynomial does more than compute a number: its simultaneous zeros carve out a geometric object.

$$V(f_1, \dots, f_r) = \{p = (a_1, \dots, a_n) \in \mathbb{A}_k^n : f_i(p) = 0 \text{ for } 1 \leq i \leq r\},$$

$$I(X) = \{f \in k[x_1, \dots, x_n] : f(p) = 0 \text{ for every } p \in X\}.$$

Line

$$V(y - x)$$



$t \mapsto (t, t)$ gives one free parameter, so the line has dimension 1.

Parabola

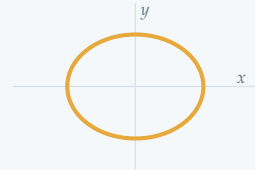
$$V(y - x^2)$$



$t \mapsto (t, t^2)$ shows that a nonlinear picture may still have one parameter.

Circle

$$V(x^2 + y^2 - 1)$$



One equation in two coordinates usually leaves one degree of freedom.

THE FIELD CHANGES THE GEOMETRY

$$V_{\mathbb{R}}(x^2 + y^2 + 1) = \emptyset, \quad V_{\mathbb{C}}(x^2 + y^2 + 1) \neq \emptyset.$$

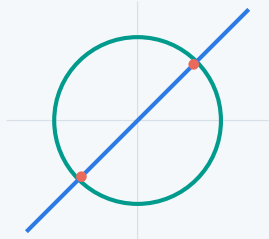
We draw real loci for intuition. The clean ideal-zero-set correspondence is strongest over an algebraically closed field, usually $k = \mathbb{C}$.

First habit. Given equations, name the ambient space, the base field, and the common zero set before trying to picture it.

Systems and ideals

A system means simultaneous vanishing. Geometrically, this is intersection. Algebraically, every polynomial consequence is packaged into an ideal.

Solve a circle-line intersection



$$\begin{aligned} x^2 + y^2 - 1 &= 0, & y - x &= 0, \\ 2x^2 &= 1, & y &= x, \\ (x, y) &= \left(\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}} \right). \end{aligned}$$

The signs are paired: the intersection has exactly two points.

$$I = (f_1, \dots, f_r) = \left\{ \sum_{i=1}^r h_i f_i : h_i \in k[x_1, \dots, x_n] \right\}.$$

Every element of I vanishes wherever $f_1, \dots, f_r$ vanish.

$$V(I + J) = V(I) \cap V(J),$$

$$V(IJ) = V(I) \cup V(J),$$

$$I(X \cup Y) = I(X) \cap I(Y).$$

More equations produce fewer points. More geometric points leave fewer polynomials vanishing everywhere.

THE ZERO SET FORGETS POWERS

$$V(x) = V(x^2), \quad (x) \neq (x^2).$$

The radical removes multiplicity that the underlying point set cannot see:

$$\sqrt{I} = \{g : g^m \in I \text{ for some } m \geq 1\}.$$

Over an algebraically closed field, Hilbert's Nullstellensatz closes the dictionary:

$$I(V(I)) = \sqrt{I}.$$

Two quick decompositions

$$V(xy) = V(x) \cup V(y),$$

$$V((y-x)(y+x)) = V(y-x) \cup V(y+x).$$

A reducible polynomial often reveals a reducible geometric set.

Local geometry and singularities

At a smooth point, the first-order terms capture the local shape. At a singular point, the linear approximation loses rank and higher-order terms become visible.

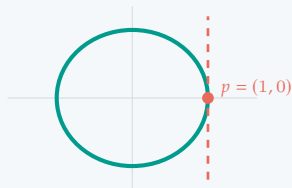
For $F = (f_1, \dots, f_r) : \mathbb{A}^n \rightarrow \mathbb{A}^r$, define

$$J_F(p) = \left(\frac{\partial f_i}{\partial x_j}(p) \right)_{i,j}, \quad T_p V = \ker J_F(p), \quad \dim T_p V = n - \text{rank } J_F(p).$$

For a hypersurface $V(f)$, smoothness at p is the condition $\nabla f(p) \neq 0$.

Smooth point on a circle

$$f = x^2 + y^2 - 1$$

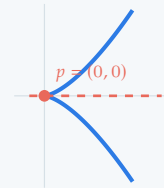


$$\begin{aligned} \nabla f &= (2x, 2y), & \nabla f(1,0) &= (2,0), \\ T_p V : 2(x-1) &= 0, & x &= 1. \end{aligned}$$

The gradient is normal to the tangent line.

Singular cusp

$$f = y^2 - x^3$$



$$\begin{aligned} \nabla f &= (-3x^2, 2y), & \nabla f(0,0) &= (0,0), \\ T_p V &= \mathbb{A}^2, & \dim T_p V &= 2 > \dim V. \end{aligned}$$

The lowest-degree term is y^2 , giving the doubled tangent direction $y = 0$.

EXPECTED RANK IS THE PRACTICAL TEST

If r independent equations cut out a local piece of codimension r , then a smooth point satisfies

$$\text{rank } J_F(p) = r, \quad \dim T_p V = n - r.$$

A rank drop signals a singularity, a component crossing, or a degeneration that deserves closer inspection.

The coordinate ring

Two polynomials define the same function on a variety when their difference vanishes there. Quotienting by those relations produces the ring of polynomial functions on the shape.

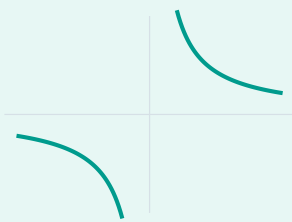
$$k[V] = \frac{k[x_1, \dots, x_n]}{I(V)},$$

$$f|_V = g|_V \iff f - g \in I(V).$$

The class $[f] \in k[V]$ is the polynomial function induced by f on V .

GEOMETRY		ALGEBRA
irreducible variety V	$\longrightarrow$	domain $k[V]$
point $p \in V$	$\longrightarrow$	maximal ideal $\mathfrak{m}_p$
closed subvariety $W \subseteq V$	$\longrightarrow$	radical ideal $I(W) \subseteq k[V]$
regular map $V \rightarrow W$	$\longrightarrow$	pullback $k[W] \rightarrow k[V]$

EXAMPLE: THE HYPERBOLA



Let $H = V(xy - 1)$. In its coordinate ring, $[x][y] = 1$, so $[x]$ is invertible:

$$k[H] = \frac{k[x, y]}{(xy - 1)} \cong k[x, x^{-1}],$$

$$[x] \mapsto x,$$

$$[y] \mapsto x^{-1}.$$

The equation has become an algebraic rule: x never vanishes on H .

For irreducible V , the coordinate ring is a domain and $\dim V = \text{trdeg}_k \text{Frac}(k[V])$, the number of algebraically independent rational parameters.

Maps between varieties

A regular map moves points forward by polynomial formulas. Functions move backward by composition. This contravariance is the organizing rhythm of the subject.

$$\begin{aligned}\varphi : V &\longrightarrow W, \\ \varphi^* : k[W] &\longrightarrow k[V], \\ (\psi \circ \varphi)^* &= \varphi^* \circ \psi^*.\end{aligned}$$

$$\begin{aligned}p &\longmapsto (\varphi_1(p), \dots, \varphi_m(p)), \\ h &\longmapsto h \circ \varphi,\end{aligned}$$

THE PARABOLA IS THE AFFINE LINE

$$\varphi : \mathbb{A}^1 \rightarrow V(y - x^2), \quad t \mapsto (t, t^2).$$

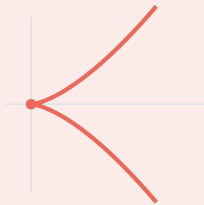
The pullback is an isomorphism:

$$\varphi^* : \frac{k[x, y]}{(y - x^2)} \xrightarrow{\sim} k[t], \quad [x] \mapsto t, [y] \mapsto t^2.$$

Its inverse sends $t \mapsto [x]$.



THE CUSP TRAP



The map $v(t) = (t^2, t^3)$ is bijective onto $C = V(y^2 - x^3)$, but

$$v^* : \frac{k[x, y]}{(y^2 - x^3)} \hookrightarrow k[t], \quad [x] \mapsto t^2, [y] \mapsto t^3$$

has image $k[t^2, t^3] \neq k[t]$. The missing function t would equal y/x away from the origin, but y/x is not regular at the cusp. Thus v is the **normalization**, not an isomorphism.

Why projective space appears

Affine space hides limiting directions. Projective space adds them and makes polynomial geometry more complete.

$$\mathbb{P}^n(k) = \frac{k^{n+1} \setminus \{0\}}{k^\times}, \quad [X_0 : \cdots : X_n] = [\lambda X_0 : \cdots : \lambda X_n] \quad (\lambda \neq 0).$$

Projective varieties are cut out by homogeneous polynomials, whose monomials all have the same total degree.

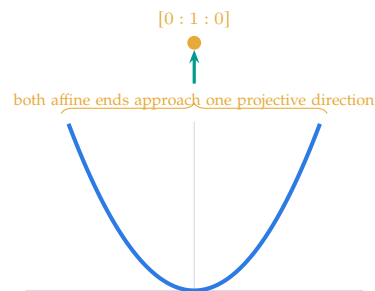
HOMOGENIZE THE PARABOLA

In the affine chart $Z \neq 0$, write $x = X/Z$ and $y = Y/Z$:

$$\begin{aligned} y - x^2 &= 0, \\ \frac{Y}{Z} - \frac{X^2}{Z^2} &= 0, \\ YZ - X^2 &= 0. \end{aligned}$$

At infinity, $Z = 0$, so $X = 0$. The projective closure gains one point:

$$\overline{V(y - x^2)} \cap \{Z = 0\} = \{[0 : 1 : 0]\}.$$



A second example

$$\begin{aligned} xy - 1 &\rightsquigarrow XY - Z^2, \\ Z = 0 &\implies XY = 0. \end{aligned}$$

The hyperbola gains two points at infinity: $[1 : 0 : 0]$ and $[0 : 1 : 0]$.

WHY THE COMPLETION MATTERS

Intersections

Curves that appear parallel in an affine chart can meet at infinity.

Completeness

Projective varieties behave like compact spaces in algebraic geometry.

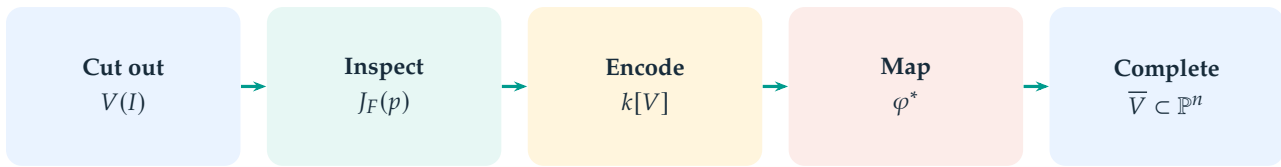
Degree

Homogeneous equations make intersection counts and scaling symmetries visible.

For example, Bezout's theorem predicts that two projective plane curves of degrees d and e , with no common component, meet in de points counted with multiplicity over an algebraically closed field.

The whole picture

Algebraic geometry is a method: cut out a space, inspect its local geometry, encode its functions, study its maps, and complete it when infinity matters.



THREE WORDS TO KEEP

$\dim V$ = number of independent local parameters,
 V irreducible $\iff V \neq X \cup Y$ for proper closed $X, Y \iff k[V]$ is a domain,
 variety = an irreducible algebraic set in the convention used here.

Some texts allow reducible varieties and reserve *integral variety* for the irreducible, reduced case.

Four quick checks

1. Decompose $V(xy) \subset \mathbb{A}^2$.

$V(xy) = V(x) \cup V(y)$,
 the two coordinate axes.

2. Find the singular locus of $y^2 = x^3$.

$$(-3x^2, 2y) = (0, 0) \implies (x, y) = (0, 0).$$

3. Homogenize $xy = 1$.

$$XY = Z^2, \quad Z = 0 \implies [1 : 0 : 0], [0 : 1 : 0].$$

4. Invert $t \mapsto (t, t^2)$.

$(x, y) \mapsto x$,
 a regular inverse on $V(y - x^2)$.

The compact dictionary $I \mapsto V(I), X \mapsto I(X), V \mapsto k[V], \varphi \mapsto \varphi^*, f \mapsto f^h$.